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Nonpharmacologic Prevention and Treatment of Hypertension

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Diets that include increased fat and refined carbohydrate intake combined with reduced physical activity have resulted in a pandemic of obesity, type 2 diabetes mellitus, and hypertension that is most pronounced in underserved and indigenous populations. Adoption of healthy lifestyles is critical in preventing and managing high blood pressure (BP). According to the American College of Cardiology/American Heart Association (ACC/AHA)[1] and International Society of Hypertension (ISH) Global Practice Guidelines, lifestyle interventions lower BP,[2] enhance efficacy of antihypertensive medication, and reduce overall cardiovascular (CV) risk.[3] The lifestyle changes that are widely agreed to lower BP and CV risk are smoking cessation, weight reduction, moderation of alcohol intake, increased physical exercise, reduced salt intake, increased fruit and vegetable intake, and decreased total fat, trans fat, and sugar intake (Table 36.1). These interventions may lower BP to a similar extent compared with single-drug therapy. However, lifestyle changes should not delay the initiation of drug therapy in patients at higher CV risk.

PREVENTION

The importance of primary prevention has been underscored by the recognition that hypertension is common, treatment is lifelong, the control of BP in hypertensive individuals does not restore CV risk to normal, and the majority of hypertensive individuals do not reach goal BP readings. The most important individuals to target are those with elevated BP (120–129/80–89 mm Hg)[4] or high BP (>130/90 mm Hg) as defined by the ACC/AHA guidelines.[1] Those with elevated BP have increased prevalence of early vascular damage, increased risk for incident hypertension, and increased risk for CV events compared with those who have optimal BP levels (<120/80 mm Hg).[5] The World Health Organization (WHO) Global Non-Communicable Disease (NCD) Alliance Action Plan advocates the following voluntary global targets for country member states: 10% reduction in harmful use of alcohol, 10% reduction in sedentary lifestyle, 30% reduction in mean population intake of salt, and 30% reduction in current tobacco use. Up to 80% of cardiovascular disease (CVD) can be prevented through lifestyle measures that include maintenance of healthy weight, adequate physical activity, a healthy diet, and avoidance of tobacco.[6] These targets align with the *Lancet*'s Commission on Hypertension (2016) that identifies key actions to prevent elevated BP at both the population and individual level.[7] A life-course approach is recommended, which focuses on early lifetime programming. Exposure to CV risk factors in early childhood has been shown to promote adverse vascular damage in early adulthood and increase the trajectory of vascular aging.[8]

WEIGHT LOSS

Obesity is epidemic throughout the world; for example, 65% of the adult population in the United States is either overweight, with a body mass index (BMI) of 25.0 to 29.9 kg/m^2, or obese, with a BMI at or above 30 kg/m^2. Obese individuals have a threefold increase in the prevalence of hypertension. Possible mechanisms for obesity-induced hypertension include overactivity of the sympathetic nervous system (SNS), hyperinsulinemia (which may increase renal sodium reabsorption), increased leptin, hyperuricemia, activation of the renin-angiotensin system, and sleep apnea. Visceral obesity is a greater predictor of both hypertension and CV risk than other types of body fat distribution. *Visceral obesity* is defined by waist circumference greater than 88 cm (>35 in) in women and greater than 102 cm (>40 in) in men. These reference values were developed in White populations and may need to be modified for other ethnic groups.

In obese hypertensive patients or those with elevated BP, weight loss of as little as 4 to 5 kg (9–11 lb) is often associated with a significant reduction in BP. A meta-analysis has demonstrated that a weight reduction of 5.1 kg reduced systolic BP (SBP) by 4.4 mm Hg and diastolic BP (DBP) by 3.6 mm Hg.[9] A rule of thumb is that for every kilogram lost, there is a reduction of 1 mm Hg in SBP and DBP. To minimize the risk for relapse and maintain sustainability of the weight loss program, the initial target should be 5% to 10% of current weight, or 1 to 2 BMI units. Marked oscillations in weight (weight cycling or yo-yo dieting) should be avoided because this increases the risk for development of hypertension in obese, normotensive persons.[10] A randomized trial of the effectiveness of four popular diets on sustained weight loss and CV disease risk reduction concluded that a variety of diets can similarly reduce weight and BP, but only a minority of individuals can maintain high dietary adherence for prolonged periods.[11]

Very-low-carbohydrate diets, such as the Atkins diet and others with a carbohydrate content below 20% of energy, are not recommended because they result in a greater increase in low-density lipoprotein cholesterol, despite resulting in a greater weight loss over the short term, compared with a traditional low-fat diet.[12] The pan-European Diet, Obesity and Genes (DiOGenes) study found that BP reduction after weight loss is better maintained when the intake of protein is increased at the expense of carbohydrates, to around 23% to 28% energy from protein, compared with usual lower protein intakes of between 10% and 15% energy.[13]

Weight reduction should be accompanied by recommendations to increase physical activity unless it is contraindicated. Bariatric surgery and pharmacologic interventions for weight loss may be useful to achieve weight loss in some patients, but always as an adjunct rather than a substitute for lifestyle modification.

TABLE 36.1 Lifestyle Modifications for Prevention and Management of Hypertension

COR	LOE	Recommendations for Nonpharmacologic Intervention
1	A	Weight loss is recommended to reduce BP in adults with elevated BP or hypertension who are overweight or obese
1	A	A heart-healthy diet, such as the DASH diet, that facilitates achieving a desirable weight is recommended for adults with elevated BP or hypertension
1	A	Sodium reduction is recommended for adults with elevated BP or hypertension
1	A	Potassium supplementation, preferably in dietary modification, is recommended for adults with elevated BP or hypertension unless contraindicated by the presence of CKD or use of drugs that reduce potassium excretion
1	A	Increased physical activity with a structured exercise program is recommended for adults with elevated BP or hypertension
1	A	Adult men and women with elevated BP or hypertension who currently consume alcohol should be advised to drink no more than two and one standard drinks[a] per day, respectively
1	A	Complete cessation of smoking

[a]In the United States, one standard drink contains roughly 14 g of pure alcohol, which is typically found in 360 mL (12 fl oz) of regular beer (usually about 5% alcohol), 150 mL (5 fl oz) of wine (usually about 12% alcohol), and 45 mL (1.5 fl oz) of distilled spirits (usually about 40% alcohol).
BP, Blood pressure; *CKD,* chronic kidney disease; *COR,* class (strength) of recommendation; *DASH,* Dietary Approaches to Stop Hypertension; *LOE,* level of evidence.
Modified from Whelton PK, Carey RM, Aronow WS, et al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults: executive summary: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *Hypertension.* 2018;71(6):1269-1324.

PHYSICAL ACTIVITY AND EXERCISE TRAINING

Physical and Sedentary Activity and Risk Mitigation for Hypertension

Physical inactivity may account for 5% to 13% of the risk for developing hypertension.[14] In addition, physical inactivity accounts for 5.5% to 25.1% of the population attributable risk for coronary heart disease. Regular physical activity and reduced sedentary activity lowers all-cause morbidity and mortality and provides the basis for public health recommendations. The 2020 WHO guidelines suggest that all adults should undertake, per week, 150 to 300 minutes of moderate-intensity, or 75 to 150 minutes of vigorous-intensity physical activity, or some equivalent combination of moderate-intensity and vigorous-intensity aerobic physical activity. The guidelines recommend regular muscle-strengthening activity for all age groups. Additionally, reducing sedentary behaviors is recommended across all age groups and has an important preventive effect with respect to hypertension and gestational hypertension.[15]

A recent review reported that patients with high BP who participated in any level of physical activity had a lower risk (by 16%–67%) of CV mortality, whereas a greater than twofold increase in CV mortality risk was observed in sedentary individuals.[16]

Effects of Exercise Training in Reduction of Blood Pressure in Normotensive and Hypertensive Individuals

In a meta-analysis of studies involving more than 1500 patients, exercise training in normotensive individuals has been shown to reduce SBP and DBP by 3.0 ± 1 and 1.7 ± 1 mm Hg, respectively.[17] However, in hypertensive patients, the effect of exercise training is even more marked. In a meta-analysis involving 27 randomized controlled trials (RCTs) and 1480 patients, aerobic activity was shown to reduce BP on average by 10.8 ± 4.7 mm Hg in hypertensive patients.[18]

Historically, it was accepted that only aerobic (dynamic) exercise was effective in lowering BP in hypertensive individuals. Other forms of exercise training, including resistance exercise, isometric resistance training, and high-intensity interval training (HIIT), are also considered safe, and they are recommended as effective exercise modalities.[1,19] However, there is no consensus whether HIIT exercise is superior to low-intensity continuous aerobic training for lowering BP.[20,21] There also appears to be benefit from a wide range of different exercise modalities, including tai chi[22] (which lowered BP by 6.58/0.57 mm Hg in a meta-analysis), qigong,[23] strength training,[24] and yoga.[25] Regular exercise also prevents the development of left ventricular hypertrophy that is independent of BP in young patients with stage 1 hypertension.[26]

Some studies indicate that mild to moderate exercise intensity (40%–70% of maximal age-predicted heart rate) is effective for lowering BP with little additional benefit associated with further increases in intensity.[17] However, exercise of higher intensity (75% maximum) is associated with a more marked and prolonged reduction in postexercise BP compared with lower-intensity exercise (50% maximum).[27] Other studies have indicated that HIIT also effectively reduces BP in hypertensive individuals. Thus, limitations in the literature remain regarding the effects of different intensity and duration of exercise on BP. Available studies are hampered by varying methodological quality, selective reporting of BP outcomes, and few participants with diagnosed hypertension.[28]

Monitoring of Blood Pressure During Exercise

A standardized approach to monitoring BP before and during exercise should be considered for hypertensive patients. If resting BP is poorly controlled, exercise training should be postponed (SBP >180 mm Hg or DBP> 110 mm Hg) until control is improved. Furthermore, if SBP rises to more than 250 mm Hg and/or DBP to more than 115 mm Hg during exercise, the training session should be terminated.[29]

Mechanisms of Blood Pressure–Lowering Effects of Exercise

The mechanism(s) involved with reducing BP in hypertensive individuals are multiple and complex. The reduction in BP immediately after exercise has been linked to a sympathetic inhibition and increased release of vasodilator substances. The mechanisms by which exercise lowers BP over the longer term are less well understood, but possibilities include reductions in systemic vascular resistance secondary to neurohumoral and structural adaptations. It has further been suggested that physical inactivity negatively affects brain areas associated with sympathetic outflow. SNS overdrive is thought to account for more than 50% of all cases of hypertension, and a lack of balance between parasympathetic and sympathetic modulation has been observed in hypertensive subjects.[30] In a recent meta-analysis of 14 trials, exercise improved the arterial stiffness that is associated with hypertension, particularly aerobic exercise, isometric exercise (static muscle contraction), and a

BOX 36.1 Practical Guidelines for Exercise in Patients With Hypertension

All apparently healthy individuals should undergo preexercise screening to determine health risk status. The American College of Sports Medicine (ACSM) recognizes that two or more of the following risk factors increase the risk associated with exercise, and individuals should undergo preexercise graded exercise testing. Risk factors include male sex (>45 years) or female sex (>55 years), serum cholesterol concentrations greater than 5.2 mmol/L, impaired glucose tolerance or diabetes mellitus, smoking, obesity (BMI ≥30), inactivity, and family history of cardiovascular disease.

Patients with uncontrolled hypertension should embark on exercise training only after evaluation and initiation of therapy. Furthermore, patients should not participate in an exercise training session if resting systolic blood pressure is above 200 mm Hg or diastolic blood pressure is above 115 mm Hg.

Many patients with hypertension are overweight and should therefore be encouraged to follow a program that combines both exercise training and restricted calorie intake.

- *Type* of exercise: this should be predominantly endurance physical activity, including walking, jogging, cycling, swimming, or dancing. This should be supplemented by resistance exercise that can be prescribed according to the ACSM or American Heart Association guidelines.
- *Frequency* of exercise: most or preferably every day, but at least 5 days a week.
- *Intensity* of exercise: moderate intensity at 40% to 60% of maximal oxygen consumption ($\dot{V}O_2$ peak).
- *Duration* of exercise: more than 30 minutes of continuous or accumulated moderate physical activity daily.

Modified from Pescatello LS, MacDonald HV, Ash GI, et al., eds. Assessing the existing professional exercise recommendations for hypertension: a review and recommendations for future research priorities. Mayo Clinic Proceedings. Elsevier; 2015; and Leosco D, Parisi V, Femminella GD, et al. Effects of exercise training on cardiovascular adrenergic system. *Front Physiol.* 2013;4:348.

combination of the two.[31] Long-term exercise training is also associated with weight loss and a reduction of serum uric acid levels, both of which could reduce BP.

Antihypertensive Medication and Guidelines for Exercise

Box 36.1 provides exercise guidelines for patients with hypertension.[32] β-Blockers decrease exercise tolerance. β-Blockers and diuretics also may alter thermoregulation in hot environments and provoke hypoglycemia. Patients using these medications should be educated about exercising in the heat, proper clothing, adequate hydration, and methods to prevent hypoglycemia.[33] However, β-blockers are not the first-line treatment for physically active hypertensive patients. Angiotensin-converting enzyme (ACE) inhibitors, angiotensin receptor blockers, and calcium channel blockers may be better suited for patients who exercise frequently or athletes with hypertension.

For patients undergoing supervised exercise training, the monitoring of postexercise BP may be helpful, so medications may be adjusted to avoid postexercise hypotension, especially with calcium channel blockers or in patients exercising in a hot environment.

DIET

Salt Intake

The prevalence of hypertension is directly related to dietary salt intake in all societies in which Na intake exceeds 50 to 100 mmol/day (3–6 g

Salt Intake and Hypertension

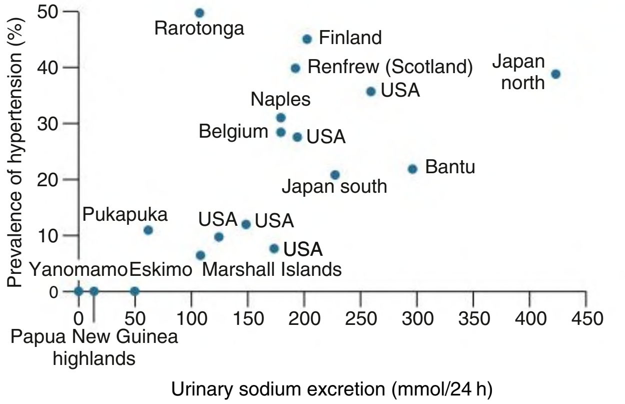

Fig. 36.1 **Relationship of Salt Intake to Prevalence of Hypertension in Different Populations.** (Modified from MacGregor GA. Sodium is more important than calcium in essential hypertension. *Hypertension.* 1985;7[4]:628–640.)

Blood Pressure Changes With Age and Salt Intake

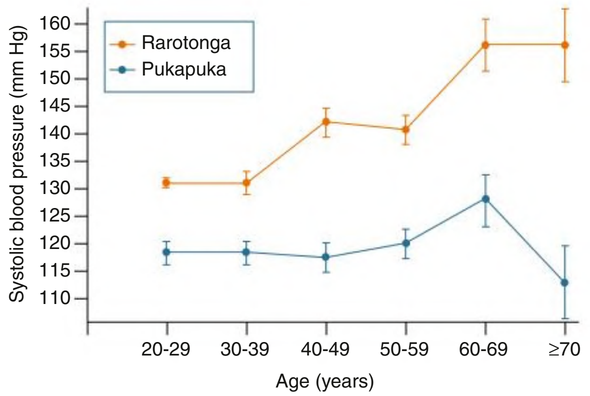

Fig. 36.2 **Blood Pressure Changes With Age and Salt Intake.** The increase in systolic blood pressure (SBP) with age correlates with a higher salt intake in two Polynesian populations. In men of Rarotonga Island, where the sodium intake averages 130 mmol/day, SBP increases with age. In contrast, it remains constant in men of Pukapuka Island, where the sodium intake averages 50 to 70 mmol/day. (Modified from Prior I, Evans JG, Harvey H, Davidson F, Lindsey M. Sodium intake and blood pressure in two Polynesian populations. *N Engl J Med.* 1968;279[10]:515–520.)

salt or NaCl) (Fig. 36.1).[34] In societies in which daily intake is below that range, hypertension is rare. Salt intake also plays an important role in age-related increase in BP (Fig. 36.2).[35] However, not all individuals respond similarly to a high salt intake. "Salt sensitivity" describes a group of individuals who significantly decrease or increase their BP during periods of salt restriction or salt loading, respectively. Those at greatest risk of salt sensitivity include Black ethnicity, older age, obese, type 1 or 2 diabetes, treatment with calcineurin inhibitors, and chronic kidney disease (CKD).

Putative mechanisms for salt sensitivity are alterations in circulating levels of (or renal responses to) atrial natriuretic factor,

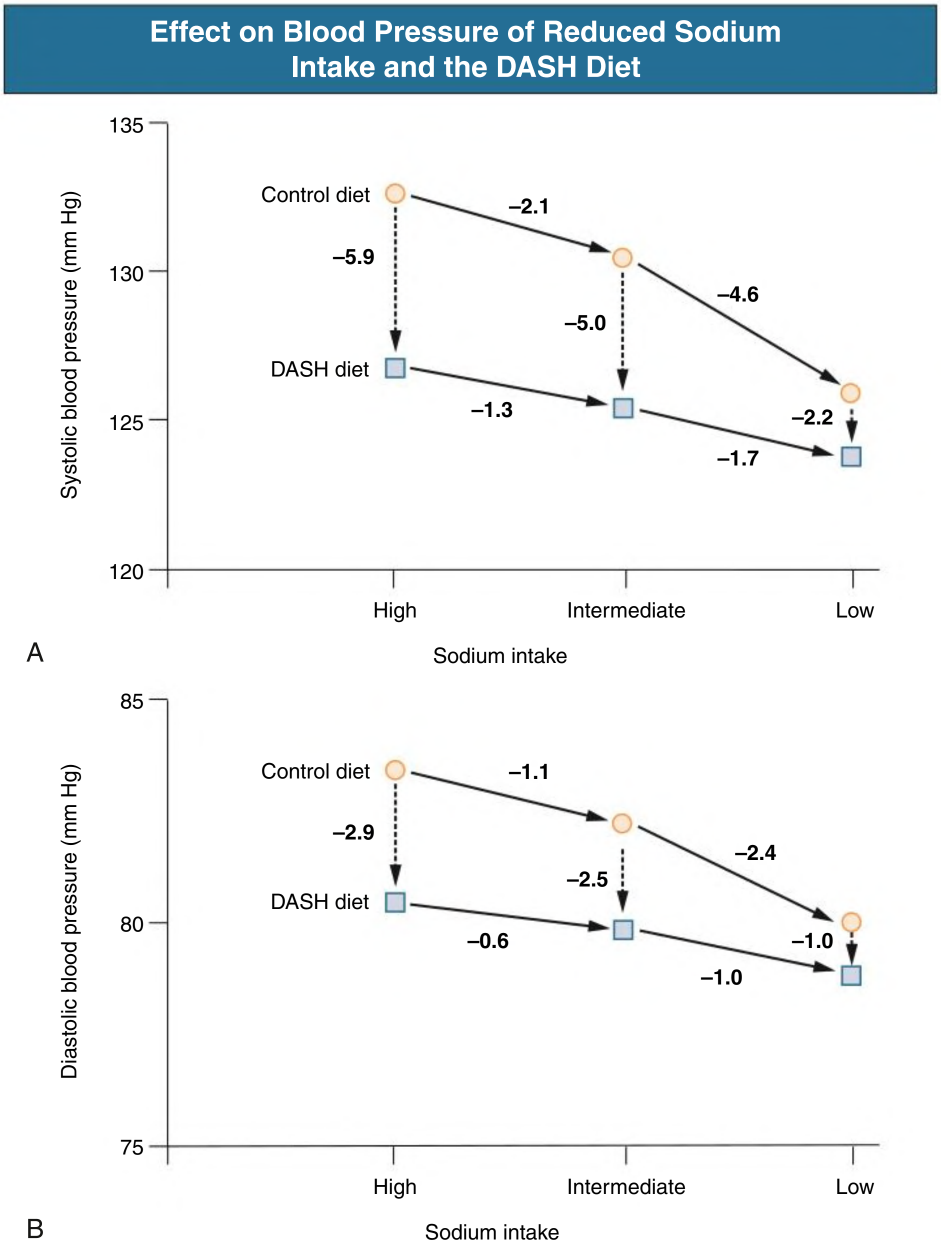

Fig. 36.3 Reduced Sodium Intake and Dietary Approaches to Stop Hypertension (DASH) Diet. Effect on systolic blood pressure (A) and diastolic blood pressure (B). (B, Modified from Sacks FM, Svetkey LP, Vollmer WM, et al. Effects on blood pressure of reduced dietary sodium and the Dietary Approaches to Stop Hypertension (DASH) diet. *N Engl J Med.* 2001;344[1]:3–10.)

kallikrein, prostaglandins, and nitric oxide (NO); increased levels of norepinephrine; abnormal suppression of both renin and aldosterone; genetic mechanisms; congenital reduction in nephron number; and acquired renal microvascular and tubular injury. More recent studies show that sodium-dependent serum osmolarity may have a higher correlation with BP[36] and the progression to hypertension.[37]

In a Cochrane systematic review of 34 trials, the effect of modest salt reduction on BP was studied. The pooled mean change in urinary sodium (Na) was −75 mmol/24 h (equivalent to reduction of 4.4 g of salt) resulting in a mean reduction in SBP of −4.18 mm Hg and DBP of −2.06 mm Hg, respectively. Meta-regression analysis showed that older age, Black ethnicity, hypertensive status, and overall change in 24-hour urinary Na were associated with a greater fall in BP.[38]

In the Dietary Approaches to Stop Hypertension (DASH) sodium trial, the additional benefits of salt restriction over and above the DASH diet were investigated (Fig. 36.3).[39] Reduction of sodium intake from high intake (150 mmol/day, or 9 g salt) to either intermediate (100 mmol/day, or 6 g salt) or low (65 mmol/day, or 4 g salt) intake resulted in a stepwise reduction in BP, which was approximately twice as great in the control group than those following the DASH diet (see Fig. 36.3). In those following the DASH diet, the addition of salt restriction resulted in a relatively small additional decrease in BP (3.0 and 1.6 mm Hg for SBP and DBP, respectively). Thus, the greatest benefits of salt restriction are seen in those with the typical Westernized high-fat, low-nutrient diet.

Most hypertension guidelines and the WHO now recommend a reduction of salt intake to less than 5 g salt (90 mmol Na) per day.

The US Department of Agriculture and US Department of Health and Human Services Joint Dietary Guidelines for Americans call for stricter reduction in salt intake to no more than 65 mmol/day, or 4 g salt in African Americans, those older than 51 years, and patients with hypertension, diabetes mellitus, or CKD.[40]

In countries where population-based reduction in salt intake has occurred (Finland, United Kingdom, and Japan), there has been an accompanying fall in BP and CV mortality. A recent AHA Presidential Advisory review concluded that the evidence supporting the effectiveness of population-level salt reduction to prevent hypertension and reduce the incidence of CV disease and stroke remains robust and persuasive for policy development.[40]

In the United States, as in most high-income countries, a major challenge to salt reduction efforts is the widespread use of sodium in the food supply, with more than 75% of total sodium intake from packaged and restaurant foods.[41] One of the most cost-effective ways to lower salt intake in the general population is to reduce salt in processed foods, as shown in an experimental study in South Africa.[42] In Belgium, reduction in the salt content of bread between the mid-1960s and the early 1980s was accompanied by marked reductions in 24-hour urinary sodium excretion. However, a multipronged approach is recommended to achieve overall salt reduction, as outlined by the WHO SHAKE Technical Package for Salt Reduction,[43] that addresses five key areas for action:

- Surveillance: measure and monitor salt use in populations.
- Harness industry: promote the reformulation of foods and meals to contain less salt.
- Adopt standards for labeling and marketing: implement standards for effective and accurate labeling and marketing of food.
- Knowledge: educate and communicate to empower individuals to eat less salt.
- Environment: support settings to promote healthy eating.

The United Kingdom was the first country to set voluntary sodium reduction targets for categories of foods through its Food Standards Agency (2009), closely followed by Australia, the United States, and Canada. South Africa was the first country to adopt mandatory regulation for maximum sodium levels across a wide range of processed food categories that are major contributors to salt intake in that population; namely, bread, margarine and spreads, savory snacks, processed meats, soup powders, and stock cubes.

Potassium Intake

One of the confounding factors of the relationship of salt and BP has been the inverse relationship between the intake of salt and that of potassium. Typically, diets with a high salt content are relatively deficient in potassium (and calcium); but in persons who consume little salt, the potassium (and calcium) intake is high.

In normotensive individuals with an average potassium intake above 1.95 g/day (50 mmol/day), potassium supplementation has no significant effect on BP. However, among hypertensive patients who are potassium deficient because of diuretic treatment or low potassium intake, potassium supplementation lowers BP (Fig. 36.4).[44] The DASH diet lowers BP and is high in potassium because of the high fruit and vegetable content and inclusion of low-fat dairy products (Table 36.2). However, the synergistic effect of the various food groups in the DASH diet makes it difficult to ascertain the contribution of the individual nutritional components. The mechanisms by which a low-potassium diet may contribute to hypertension are poorly understood but may relate to stimulation of intrarenal angiotensin II (Ang II), oxidants, and endothelin; inhibition of intrarenal NO and prostaglandins; and induction of renal ischemia.

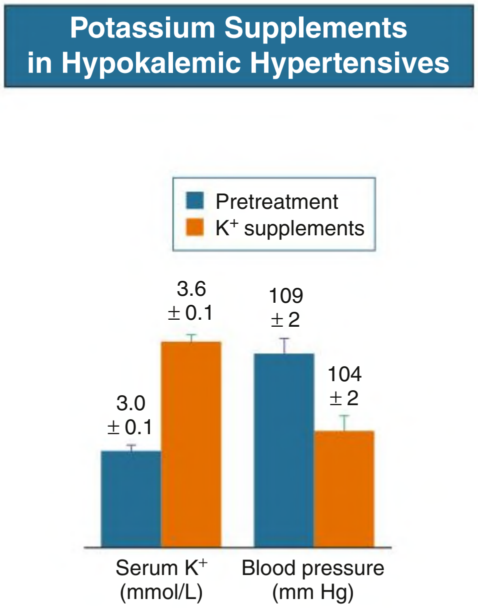

Fig. 36.4 Potassium Supplementation Lowers Blood Pressure in Hypokalemic Hypertensive Patients. Treatment with potassium chloride (60 mmol/day potassium for 6 weeks) resulted in an increase in serum potassium concentration and a decrease in mean arterial pressure in hypertensive patients taking thiazide diuretics. (Modified from Kaplan NM, Carnegie A, Raskin P, Heller JA, Simmons M. Potassium supplementation in hypertensive patients with diuretic-induced hypokalemia. *N Engl J Med.* 1985;312[12]:746–749.)

A systematic review of clinical trials identified that a lower 24h urine Na/K ratio was associated with a significantly greater reduction in both systolic and diastolic BP. Dietary strategies to achieve an increase in potassium while at the same time lowering sodium would be beneficial in lowering BP.[45] Overall, it appears to be beneficial to optimize potassium intake in hypertensive patients to minimize hypokalemia, being careful to avoid the risk for hyperkalemia, especially in those with reduced glomerular filtration rate. If kidney function is normal, optimal potassium intake is 80 to 120 mmol/day. For CVD prevention, the WHO recommends a potassium intake that will keep the urine Na/K ratio close to 1 (i.e., 70–80 mmol/day if Na guidelines are met).

Calcium, Vitamin D, and Dairy Food Intake

Cross-sectional population surveys of self-reported nutrient intake suggest an inverse relationship between calcium intake and BP. The relationship is more convincing at low levels of calcium consumption (<300–600 mg/day). There may be a threshold of 700 to 800 mg/day, above which any further reduction in BP is attenuated. A meta-analysis of randomized calcium supplementation trials (mostly with 1 or 1.5 g calcium daily) demonstrated reductions in SBP (−0.9 to −1.7 mm Hg) that are of little clinical importance.[46] Although calcium in milk may contribute to BP lowering, dairy products may lower BP by other mechanisms. Biologically active peptides formed during the milk fermentation process, such as the casein-derived tripeptides isoleucine-proline-proline and valine-proline-proline, have ACE-inhibiting properties. Vitamin D, which is often added to milk, also may help reduce BP by reducing renin expression, but in a randomized controlled clinical trial, vitamin D had no effect on BP compared with placebo.[47] At present, the AHA does not recognize dairy consumption as a dietary approach to the prevention and management of hypertension. Nevertheless, low-fat dairy products are recommended as an integral part of the DASH diet.

TABLE 36.2 Dietary Approaches to Stop Hypertension (DASH) Diet[a]

Food Group	Daily Servings	Serving Sizes	Examples and Notes	Contribution to the DASH Diet Pattern
Grains and grain products	7–8	1 slice bread ½ cup dry cereal ½ cup cooked rice, pasta, or cereal	Whole-wheat bread, muffin, pita bread, bagel, cereals, oatmeal	Major sources of calories and fiber
Vegetables	4–5	1 cup raw, leafy vegetables ½ cup cooked vegetables 6 oz vegetable juice	Tomatoes, potatoes, carrots, peas, squash, broccoli, turnip greens, kale, spinach, artichokes, green beans, sweet potatoes	Rich sources of potassium, magnesium, and fiber
Fruits	4–5	1 medium fruit ½ cup dried fruit 6 oz fruit juice ½ cup fresh, frozen, or canned fruit	Apricots, bananas, dates, oranges, orange juice, grapefruit, grapefruit juice, mangoes, melons, peaches, pineapples, prunes, raisins, strawberries, tangerines	Important sources of potassium, magnesium, and fiber
Low-fat or nonfat dairy foods	2–3	8 oz milk 1 cup yogurt 1.5 oz cheese	Skim or low-fat (2%) milk, skim or low-fat buttermilk, nonfat or low-fat yogurt, nonfat or low-fat cheeses	Major sources of calcium and protein
Meats, poultry, and fish	≤2	3 oz cooked meats, poultry, or fish	Select only lean meats; trim away visible fats; broil, roast, or boil instead of frying; remove skin from poultry	Rich sources of protein and magnesium
Nuts, seeds, and legumes	4–5/wk	1.5 oz or ½ cup nuts ½ oz or 2 tbsp seeds ½ cup cooked legumes	Almonds, mixed nuts, peanuts, walnuts, sunflower seeds, kidney beans, lentils, split peas	Rich sources of calories, magnesium, potassium, protein, and fiber

[a]The DASH eating plan shown is based on 2000 kcal/day. Depending on energy needs, the number of daily servings in a food group may vary from those listed.

Magnesium Intake, Other Micronutrients, and Bioactive Food Components

A weak inverse relationship has been reported between dietary intake of magnesium and BP, and in a meta-analysis of 34 RCTs, magnesium supplementation with a median dose of 368 mg/day was associated with a 2 mm Hg and 1.78 mm Hg reduction in SBP and DBP, respectively.[48]

In contrast, the inverse association between fruit and vegetable intake and BP and other CV risk factors is well established. Epidemiologic evidence suggests that polyphenol compounds found in fruit may partially explain the cardioprotective properties of fruits. Intervention trials have shown that fruits containing relatively high concentrations of flavonols, anthocyanins, and procyanidins, such as pomegranate, purple grapes, and berries, were effective at reducing CVD risk factors. Regular consumption of flavonol-rich foods, cocoa products, tea, and red wine may reduce BP. Folate, B vitamins, and L-arginine supplementation[49,50] may also lower BP and decrease overall CV risk. Foods that have a high content of inorganic nitrate (NO_3), such as beetroot, are considered to be a potential complementary treatment for hypertension.[51-53] Inorganic NO_3 supplementation may compensate NO-disrupted pathways in hypertension and increase NO bioavailability, an important physiologic mediator in regulating BP.[54] The latest ISH guidelines for lifestyle modification include advice for moderate intake of beetroot juice, pomegranate juice, and cocoa.[2]

Dietary Sugars and Fats

Added sweeteners, such as table sugar and high-fructose corn syrup, have been linked with the epidemics of obesity, hypertension, metabolic syndrome, diabetes, and CV disease.[55] Experimental studies suggest that fructose may increase the risk for obesity and diabetes because of its unique ability to reduce intracellular adenosine triphosphate levels and generate uric acid.[56] One study has reported that reducing soft drink intake by one drink per day is associated with a decrease of 1.8 mm Hg SBP.[57]

Interestingly, whole fruits appear to be beneficial despite containing fructose, possibly as a result of the high content of protective nutrients such as vitamin C, antioxidants, flavanols, potassium, and fiber. Possibly because of this, a systematic review and meta-analysis of three prospective cohort studies in 37,375 men and 185,855 women reported no association of fructose consumption with the incidence of hypertension.[58] However, although there was no excess risk associated with average levels of fructose consumption, a positive association was identified between the incidence of hypertension and high intake of fructose.

Supplementation with omega-3 fatty acids reduces the risk for myocardial infarction (MI) and sudden cardiac death, but their effect on BP is small. In a meta-analysis, omega-3 supplementation significantly reduced DBP by a mean of 1.8 mm Hg but had no effect on SBP, fibrinogen level, or heart rate.[59] About 10 portions of oily fish per week or 9 or 10 fish oil capsules per day are required (equivalent to ~3 g/day long-chain n-3 fatty acids), but this is not tolerated by most because of belching and fishy taste. Concerns about the cholesterol content, as well as dioxin and polychlorinated biphenyl content (environmental pollutants that have carcinogenic potential and, being fat soluble, can accumulate in the body) of some fish oil supplements also raise questions about the safety of very large doses. As a guideline for overall health, individuals with hypertension should aim to consume about 2 to 3 portions (200–400 g) of oily fish (e.g., herring, kippers, mackerel, pilchards, sardines, salmon, trout, fresh tuna, swordfish) per week.

Dietary Approaches to Lower Blood Pressure

Although individual nutrients and components of foods can have an effect on BP, it is necessary to consider these within the context of a total dietary approach because of potential synergistic effects.[60] A meta-analysis of 41 studies of plant-based diets showed that these diets lowered the SBP and DBP, irrespective of sex and BMI, by 4.29 mm Hg and 2.79 mm Hg, respectively.[61] These dietary patterns included the DASH diet, the Mediterranean diet, and the Nordic diet, and the key factor is that they all emphasize plant-based foods, rich in fruits, vegetables, whole grains, legumes, nuts, seeds, dairy, and fish and low in processed foods and red meat. Both the DASH diet and the Mediterranean diet have been shown to decrease the decline in renal function, progression to dialysis, and mortality.[62]

SMOKING

Cigarette smoking is a major risk factor for CV disease, but its specific role in the development of hypertension is not well elucidated and not routinely included in recommendations for prevention and treatment of hypertension. The relationship with hypertension may be confounded by changes in weight during and after the cessation of smoking. In a large epidemiologic study of middle-aged and older men from the United States, smoking was associated with a modest but important risk for development of hypertension.[63] Furthermore, BP is elevated immediately after using nicotine-containing products.[64] In addition, smoking increases the risk for CKD progression, as well as morbidity and mortality from multiple causes, so all smokers should be advised to stop.

ALCOHOL

There is a linear relationship between alcohol consumption, BP levels, and prevalence of hypertension. In Japan, alcohol intake above 300 g/wk (about three drinks daily) was associated with significantly greater increases of BP during a 7-year period, and baseline BP was higher in drinkers consuming 200 g/wk.[65] Heavy drinking is associated with increased risk for stroke, increase in BP after alcohol withdrawal, and attenuation of antihypertensive efficacy. Paradoxically, alcohol has a J-shaped relationship with coronary heart disease, with moderate consumption (one to two drinks daily) having the lowest risk. In a large epidemiologic study, modest alcohol consumption was protective against first MI.[66] Alcohol may increase BP through activation of the SNS, whereas its protective effects include increasing high-density lipoprotein cholesterol, lowering fibrinogen, and inhibiting platelet activation. The ACC/AHA guidelines recommend that in adult men and women with elevated BP or hypertension who currently consume alcohol should be advised to drink no more than two and one standard drinks per day, respectively (Table 36.1).[1] One standard drink is equivalent to 360 mL (12 fl oz) beer, 150 mL (5 fl oz) wine, or 45 mL (1.5 fl oz) 80-proof spirit.

CAFFEINE

Caffeine is the most widely used psychoactive substance worldwide. Caffeine stimulates the CV system through the blockade of vascular adenosine receptors. In a meta-analysis of RCTs of coffee ingestion, there was no effect on BP or risk for hypertension, although the quality of the evidence was low.[67] The ISH global guidelines recommend moderate consumption of tea or coffee, but pharmacologic ingestion of caffeine and the use of "smart drinks" supplemented with caffeine should be avoided.[2]

PSYCHOLOGICAL STRESS

Chronic psychological stress is a contributor to the development and maintenance of hypertension. Men exposed to job strain had a 10.7 mm Hg and 15.4 mm Hg higher work and home ambulatory SBP than did controls, respectively.[68] The INTERHEART study showed that psychosocial stress was associated with a twofold increase in the risk for the first MI.[66] Although stress reduction techniques may be beneficial for other reasons, there are few long-term data on efficacy in reducing BP. However, there is rising evidence that stress reduction may assist in blood BP control,[25] and it is recommended in the most recent ISH global hypertension guidelines.[2]

ADOPTING LIFESTYLE MODIFICATIONS

Maintaining adherence to lifestyle changes has always been challenging. The Prevention of Myocardial Infarction Early Remodeling (PREMIER) trial evaluated the effects of implementing JNC 7 lifestyle recommendations and the DASH diet in adults with prehypertension or untreated stage 1 hypertension.[69] At 6 months, there was little additional benefit in adding the DASH diet to JNC recommendations, but both these intervention groups had greater BP reductions compared with a general advice-only group. Importantly, participants purchased their own foods instead of being provided with foods as in the DASH and DASH low-salt diet studies. In the DASH low-salt diet study in which all foods were provided, salt reduction alone resulted in a BP-lowering effect of −6.7/−3.5 mm Hg.[39] A meta-analysis of 34 salt restriction trials in which participants mostly prepared their own low-salt meals reported an effect size of only −4.18/−2.06 mm Hg.[38] Thus, even highly motivated individuals usually cannot meet DASH dietary goals unless their meals are provided.

The TOHP II study demonstrated the problems of sustainability of dietary intervention and the need for regular counseling.[70] The effect of adding salt restriction to weight loss appeared to offer no further decrease in BP, but this was explained by poor compliance to salt reduction. Higher attendance at counseling sessions was associated with a greater reduction in urinary sodium, and the targets were met only in those participants who attended more than 80% of counseling sessions. In summary, the sustainability of long-term lifestyle interventions remains problematic, but it appears that regular and long-term counseling can improve adherence to both medication and lifestyle changes.[71]

SELF-ASSESSMENT QUESTIONS

1. In regard to exercise and hypertension, select the *incorrect* answer:
   - **A.** Physical inactivity may account for 5% to 13% of the risk for developing hypertension.
   - **B.** Regular physical activity and reduced sedentary activity lowers all-cause morbidity and mortality.
   - **C.** The 2020 World Health Organization guidelines suggest that all adults should undertake 150 to 300 minutes of moderate-intensity physical activity or 75 to 150 minutes of vigorous-intensity physical activity.
   - **D.** The guidelines do not recommend regular muscle-strengthening activity for all age groups.

2. For every kilogram reduction in weight, it is estimated that the systolic blood pressure will be reduced by:
   A. 0.5 mm Hg.
   B. 1 mm Hg.
   C. 2 mm Hg.
   D. 3 mm Hg.
3. Which of the following patients should not be targeted for stricter sodium reduction, according to the US Department of Agriculture and the US Department of Health and Human Services?
   A. African Americans
   B. Patients with chronic kidney disease
   C. Hypertensive individuals
   D. People younger than 40 years
4. In regard to alcohol consumption, which of the following statements is *incorrect?*
   A. There is a linear relationship between alcohol consumption, blood pressure levels, and prevalence of hypertension.
   B. Abstinence from alcohol may reduce the risk of myocardial infarction.
   C. Reduction of alcohol consumption to 1 to 2 standard drinks is recommended.
   D. Heavy drinking is associated with increased risk for stroke.

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